

CHAPTER 1

ANALYSIS I SEMINAR OVERVIEW

An effective Predictive Maintenance Program (PMP) is a total program of the following:

- 1. DETECTION**
- 2. ANALYSIS**
- 3. CORRECTION**
- 4. VERIFICATION**

This is a logical sequence of steps. The program first helps you detect the onset of a problem. It then provides means for analyzing the problem in order to determine its cause. It puts you in a position to correct the problem, effectively and efficiently, at a convenient time. And finally, it gives you a means to verify that any correction taken did in fact correct the problem and that no other problems were included.

Predictive Maintenance uses the process of tracking vibration levels on equipment components to determine the condition of the machinery. The guiding purpose of this seminar is to provide instruction on how vibration signature analysis can be used to continually evaluate machine condition within a Predictive Maintenance Program.

Predictive Maintenance Programs begin with Baseline (or initial) surveys of machines. Later, follow-up surveys are conducted at periodic intervals dependent on machine type, criticality, operating and maintenance cost, operating speed and design of components within such as bearing type, gearing type, etc. Follow-up either Baseline or Follow-up Periodic Survey measurements, an analysis of collected data is made and written survey reports are compiled summarizing diagnostic results as well as providing overall recommendations such as those which follow:

- 1) No Problem Found.
- 2) Minor Problems Found - Trend Only at this Time During Future PMP Surveys.
- 3) Potentially Serious Problems Detected Which Might Tend to Deteriorate Within Weeks - Continue to Monitor Equipment at Shorter Intervals.
- 4) Potentially Serious Problems Detected, but Problem Source Not Yet Confirmed - Perform Vibration Diagnostics to Determine Problem Source and Severity.
- 5) Significant Problem(s) Detected Which Warrant Corrective Action - Replace Parts at Next Scheduled Shutdown.
- 6) Very Severe Problem(s) Detected Which Mandate Corrective Action - Shut Down and Replace Immediately.

One of the overall purposes of this seminar will be to teach the student how to begin making these diagnostic calls using vibration signature analysis, how to evaluate problem severities and what thought process should be used to either recommended further diagnostic tests, or how to go ahead and recommend what corrective actions should be made in which chronological order.

Importantly, the analyst will then be instructed to verify all problems have been resolved (and no new ones have been introduced) after corrective actions have been taken. In other words, he should repeat his analysis to assess the machine condition after completion of such repairs. Documenting what actions were taken, along with the date they were completed, is an essential step to any successful program. However, taking the extra step to document the "Before" and "After" condition is what is critically needed to keep the program visible and effective in the minds of those associated, not only with Maintenance and Plant Management, but also to those in the Production Department.

Following the completion of this course, the user will have a good working knowledge of the proper application instrumentation and software necessary for setting up of and implementing an effective predictive maintenance program as well as receive instruction on basic concepts involved in troubleshooting vibration problems. The following are brief introductions for each of the chapters covered in this seminar text.

- **CHAPTER 2 - "WHAT IS VIBRATION AND HOW CAN IT BE USED TO EVALUATE MACHINE CONDITION?":** describes the basic theory of vibration which includes amplitude, frequency and phase. It also introduces the time waveform and how it is converted into a vibration spectrum. The relationships between acceleration, velocity, and displacement are illustrated, as well as their direct relation to machine wear, fatigue, and failure. The true meaning of Overall Vibration is also given here, along with how it is measured. This section also points out the great difference in whether this overall level is directly measured from a time waveform (Analog method), or if it is calculated from only the data within the spectrum itself between F_{MIN} and F_{MAX} (Digital method). It shows how much of a difference this can make simply by which method is employed. Also included are examples of trending vibration levels on equipment and determining vibration severity.
- **CHAPTER 3 - "OVERVIEW OF THE STRENGTHS AND WEAKNESSES OF TYPICAL VIBRATION INSTRUMENTS":** introduces the user to the types of vibration measurement equipment commonly found in a Predictive Maintenance Program as well as the vibration instruments and software required to analyze and diagnose vibration problems. A brief history of the development and enhancement of these tools is provided, along with updated information on some of the more state-of-the-art PMP systems available today.
- **CHAPTER 4 - "OVERVIEW OF VARIOUS VIBRATION TRANSDUCERS AND THEIR OPTIMUM APPLICATIONS":** describes the various transducers (vibration pickups) that measure the vibration oscillation and transmit this information back to the measuring instrument. Transducers are covered which can directly measure any of the three primary vibration parameters including acceleration, velocity, or displacement. Information on which particular transducer should be employed based on machine speed, machine components and overall frequency response is also covered.
- **CHAPTER 5 - "ROLE OF SPIKE ENERGY, HIGH FREQUENCY DETECTION (HFD) AND SHOCK PULSE (SPM), ALONG WITH PROPER SPECIFICATION OF THEIR ALARM LEVELS AT VARIOUS SPEEDS":** describes the theory behind each of these parameters and illustrates the types of machinery problems that are and are not responsive to these high frequency signals. Acceptable spike energy, HFD and shock pulse levels are provided from severity charts which have been developed for these parameters.

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- **CHAPTER 6 - "INTRODUCTION TO VIBRATION SIGNATURE ANALYSIS AND HOW IT IS USED TO DIAGNOSE MACHINE OPERATING CONDITION":** introduces the user to Technical Associates' renowned 5-Page "Illustrated Vibration Diagnostics Chart" which shows the user a typical spectrum for each of 48 problem conditions (unbalance, misalignment, rolling element bearing problems, etc.), demonstrates phase response for those problems which have an effect on phase, and provides supporting remarks summarizing the more important symptoms each problem will normally display. Following presentation of the diagnostic chart will be discussions separately covering many of the problem conditions listed in the chart. However, since this seminar is intended to be an introductory course, only the more common problems will be covered (all remaining chart items are covered in the next seminar module). Included in many of the theory discussions are vibration spectra taken from machines having the particular problem being covered. Finally, real-world case histories representing each of the problems covered in Chapter 6 are included in the last chapter of the seminar text.
 - **CHAPTER 7 - "PROVEN METHOD FOR SPECIFYING SPECTRAL BAND ALARM LEVELS AND FREQUENCIES USING TODAY'S PREDICTIVE MAINTENANCE SOFTWARE SYSTEMS":** illustrates how to properly specify not only overall vibration alarms for each point on each machine in a program, but also how and why various portions of each spectrum should be set apart, allowing much more for some of these so-called "spectral alarm bands" than for others. For example, if one band is set around operating speed (1X RPM), whereas another band is specified around bearing defect frequencies for the machine, much more vibration is most always allowed for 1X RPM as compared to that for bearing defect frequencies (which are covered in Chapter 6). Note in this chapter that different machine types will have a different spectral band setup (for example, a pump having a 6-vaned impeller and rolling element bearings versus a motor outfitted with sleeve bearings). These spectral bands are available in several Predictive Maintenance Software systems. The vibration amplitudes within these bands are summed to produce a so-called "power" level in each band which is compared with an allowable level for that band specified by the user. Information is given on how this so-called "RSS Power Level" is calculated for each band and why this is important when attempting to specify meaningful alarm levels for each spectral band. Details of the starting and ending frequencies for each band are discussed, along with the band alarm as per the type of equipment being monitored. Also provided is an important parameter - where the maximum frequency (F_{MAX}) should be set for each type of machine, bearing type and operating speed.
 - **CHAPTER 8 - "COMMON PITFALLS IN EVERYDAY VIBRATION MEASUREMENTS AND THEIR EFFECT ON PROPERLY DIAGNOSING MACHINE PROBLEMS":** informs the user on the proper choice and use of vibration transducers based on transducer mounting, frequency range, and output parameter (acceleration, velocity, or displacement) desired. The pros and cons of the vibration measurement location (bearing housing, shaft, etc...) are also evaluated. It also clearly demonstrates the detrimental effect on accuracy of diagnostic calls if the wrong transducer (or transducer mount) is used; if the wrong frequency span is measured; if measurements are made from the wrong location on a machine; etc.
 - **CHAPTER 9 - "SETUP AND IMPLEMENTATION OF EFFECTIVE PREDICTIVE MAINTENANCE AND CONDITION MONITORING PROGRAMS":** pulls together the information on all the PMP hardware and software tools, signature analysis and overall/spectral band alarm settings covered in the previous chapters and shows the user how to use this knowledge to put together a complete and cost effective Predictive Maintenance Program (PMP). Implementation costs vs. payback savings are presented. A flow chart describing the steps necessary to create an effective Predictive Maintenance Program is presented, as well as the necessary data sheets that are required for determining proper alarm settings, frequency ranges, measurement parameters (i.e., acceleration, velocity, displacement, etc.), as well as how to determine optimum data collection routes. Examples of machine condition reports are provided which include frequency spectra from damaged machines; overall condition for each machine evaluated;

severity rank-ordered reports for problem machines; alarm exception reports; and machine measurement reports which show the overall level measured and recorded for each point on each machine evaluated.

- **CHAPTER 10 - "ACTUAL CASE HISTORIES OF VIBRATION DIAGNOSTIC ON VARIOUS MACHINE TYPES"**: offers an array of real-world problems which have been detected by Predictive Maintenance using the tools taught in this seminar. They show how problems were determined by vibration diagnostics, and subsequently corrected without catastrophic failure. Impressive "before" and "after" frequency spectra are displayed to show the effect of completing the recommended corrective actions on the machine, and thereby prolonging the life of the equipment by reducing vibration amplitudes.

SEMINAR GOALS

At the conclusion of this seminar, it is hoped the user of this manual will be able to effectively communicate on a technical level with others in the field of vibration analysis. In addition, he should also gain a fundamental understanding of how to effectively start up and fully implement an effective Predictive Maintenance Program. Furthermore, an introduction into the use of vibration analysis for purposes of troubleshooting equipment problems will have been presented by this manual. Your questions and comments pertaining to this course's educational content are welcome and will provide the basis for enhancements to better train future seminar students.